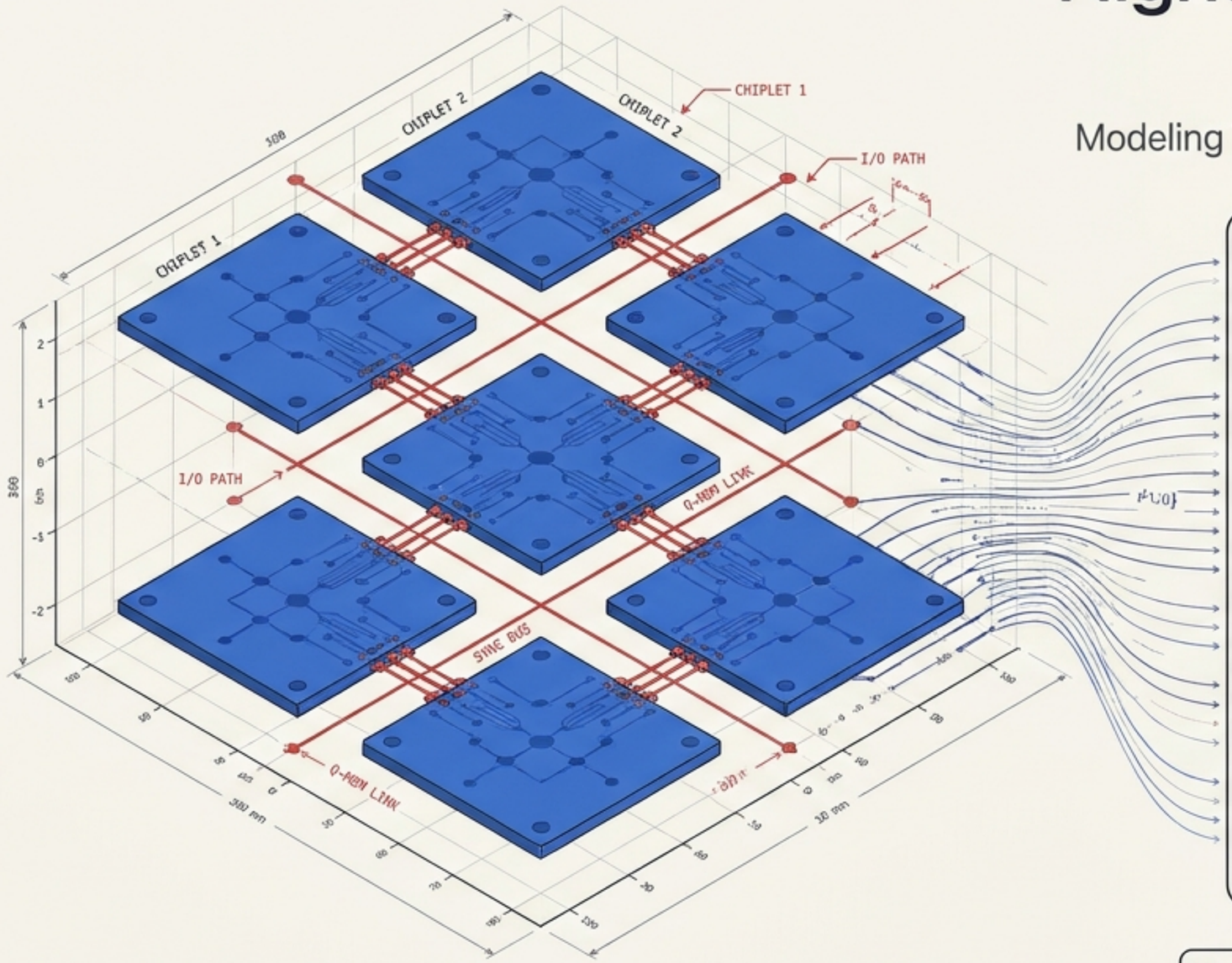
Nighthawk r3: Consolidated Quantum Simulation

Modeling 720-qubit multi-chiplet architectures on PyQrack.




```python
# Nighthawk r3 Quantum Simulation Interface
import pyqrack aa q
from nighthawk_sim import MultiChipletArchitecture, ResourceMonitor

# Initialize 2x3 Chiplet Grid
chiplet_grid = MultiChipletArchitecture(rows=2, cols=3, qubits_per_chiplet=120)
# Total Qubits: 720

# Define Classical Interconnects (Brick Red)
interconnects = {
    (0,0): [(0,1), (1,0)],
    (0,1): [(0,0), (0,2), (1,1)],
    (0,2): [(0,1), (1,2)],
    (1,0): [(0,0), (1,1)],
    (1,1): [(0,1), (1,0), (1,2)],
    (1,2): [(0,2), (1,1)]
}
chiplet_grid.set_interconnects(interconnects)

# Quantum State Initialization (Cobalt Blue)
sim = q,QrackSimulator(num_qubits=720)
sim.initialize_state('ground_state')

# Apply Gates & Measure (Burnt Orange for Constraints)
# Memory Limit Check: 8GB VRAM Constraint
if sim.estimate_memory_usage() > 8589934592: # 8GB in bytes
    raise ResourceError("Memory limit exceeded for 720-qubit simulation")

# Run Simulation & Collect Data
results = sim.run(shots=1824)
benchmark_data = chiplet_grid.analyze_performance(results)

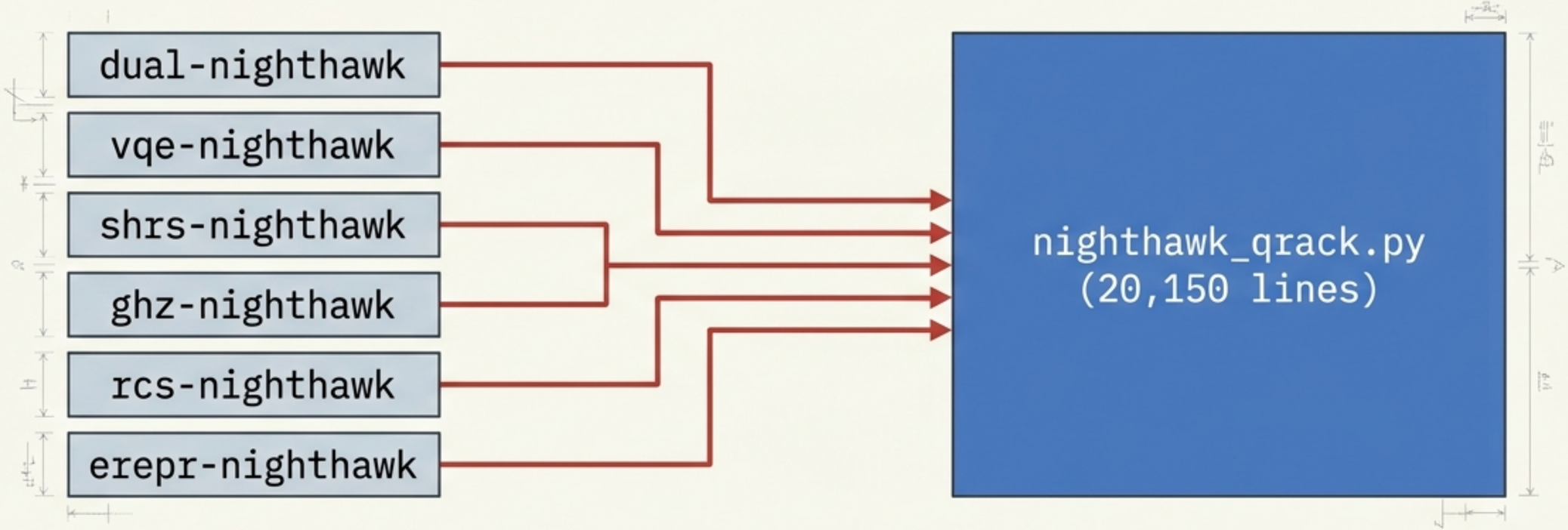
print(f"Simulation Complete. Fidelity: {benchmark_data['fidelity']:.4f}")
print(f"Memory Usage: {benchmark_data['memory_usage_gb']:.2f} GB")
```

Diagnostic Reference & Benchmark Guide | 8GB VRAM Architecture

Gemini Notebook

One Engine for Every Simulation

20,150 lines of unified Python merging seven repositories into a single executable, executed purely via PyQrack (OpenCL/CPU, no CUDA).




```
Old: nighthawk_ghz.py <cmd>          → New: nighthawk_qrack.py ghz <cmd>
Old: nighthawk_cloverfield.py <cmd> → New: nighthawk_qrack.py field <cmd>
Old: nighthawk_erepr.py <cmd>        → New: nighthawk_qrack.py erepr <cmd>
```

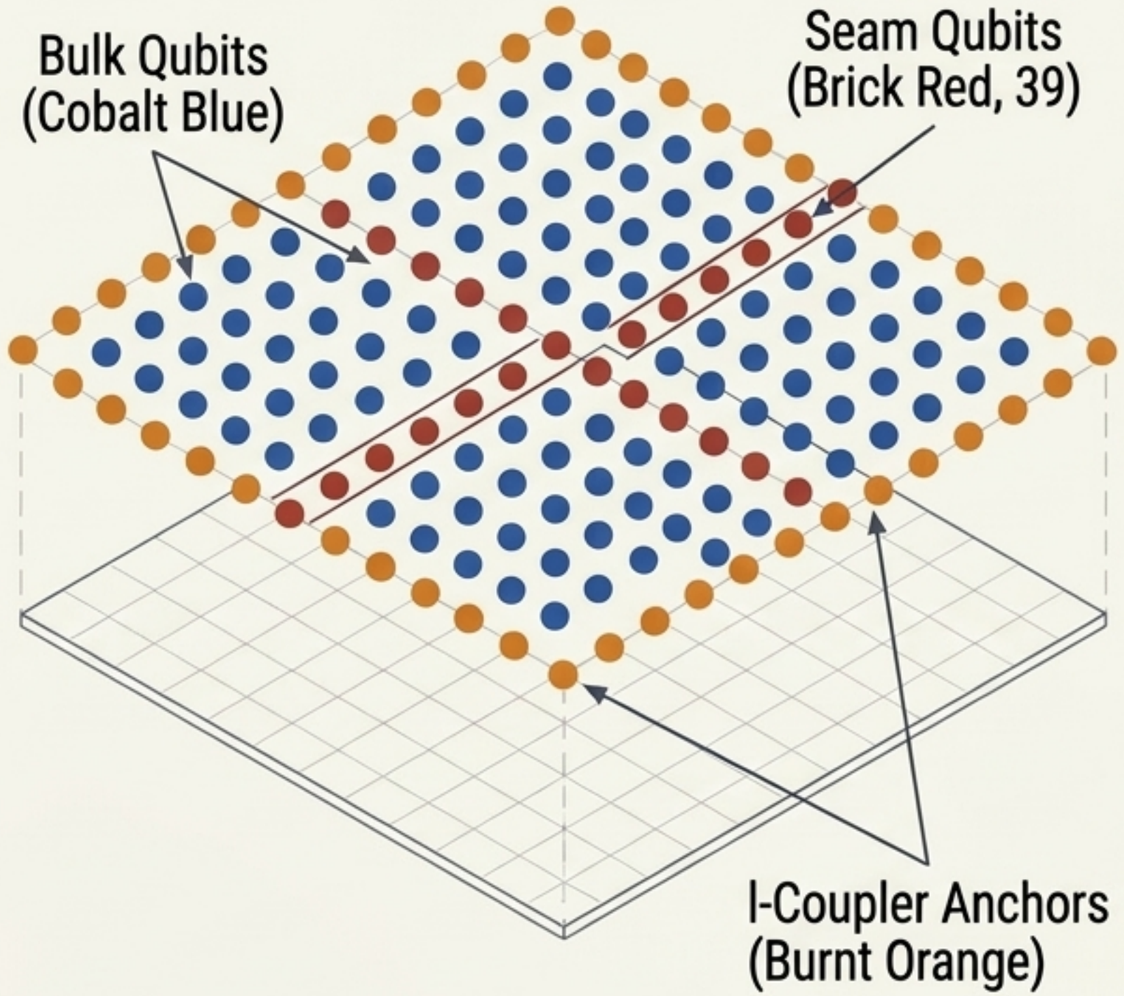
Gemini Notebook

The r3 Processor Blueprint: Target vs. Model

Target (IBM QDC 2026)	Simulation (The PyQrack r3 Model)
Qubits: 520 per processor	Qubits: **720** (six **120-qubit** chiplets)
Two-Qubit Gates: 10,000	Two-Qubit Gates: **No hard limit** (budget supports ~29 cycles of the full field)
Architecture: 2 linked processors (1,040q) by 2027	Architecture: r3x2 preset supports **1,440 qubits.**
Lattice: Square	Lattice: **12x10 square** per chiplet, 10 **l-couplers** per adjacent pair.

Gemini Notebook

Mapping the 120-Qubit Chiplet (Memory Constraints)



nighthawk120

0.3 GiB Memory Cost (Fits in 8GB VRAM easily).

184 ACE sites, **8** patches, Max patch width: **23**.

nighthawk120w

4.75 GiB Memory Cost.

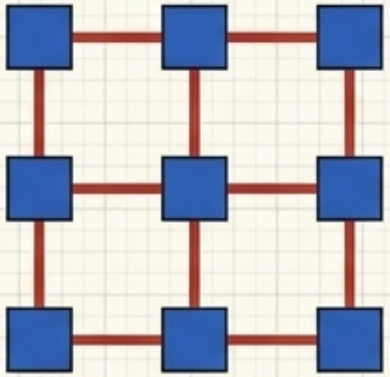
170 ACE sites, **6** patches, Max patch width: **27**.

`use_crossbars=False` and ACE error detection can return NaNs past 4 cycles. Run deeper chiplets with `--no-ace-error-detection`.

Gemini Notebook

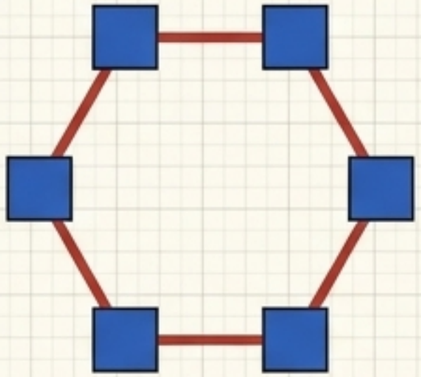
Routing the Field: Topology Models

Grid



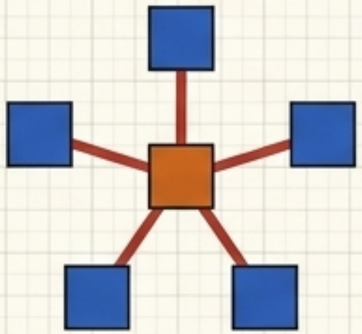
Default. 70 links. 1.67 mean hops.

Ring



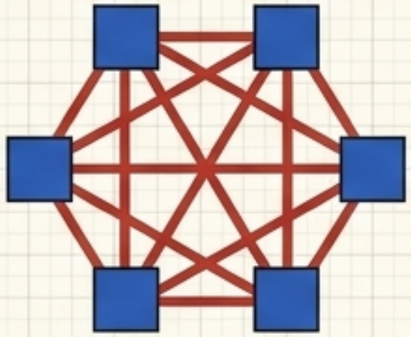
Perimeter only. 60 links. 1.80 mean hops.

Hub



Single router chip. 40 links (16 long). (Severe bottleneck).

Wheel

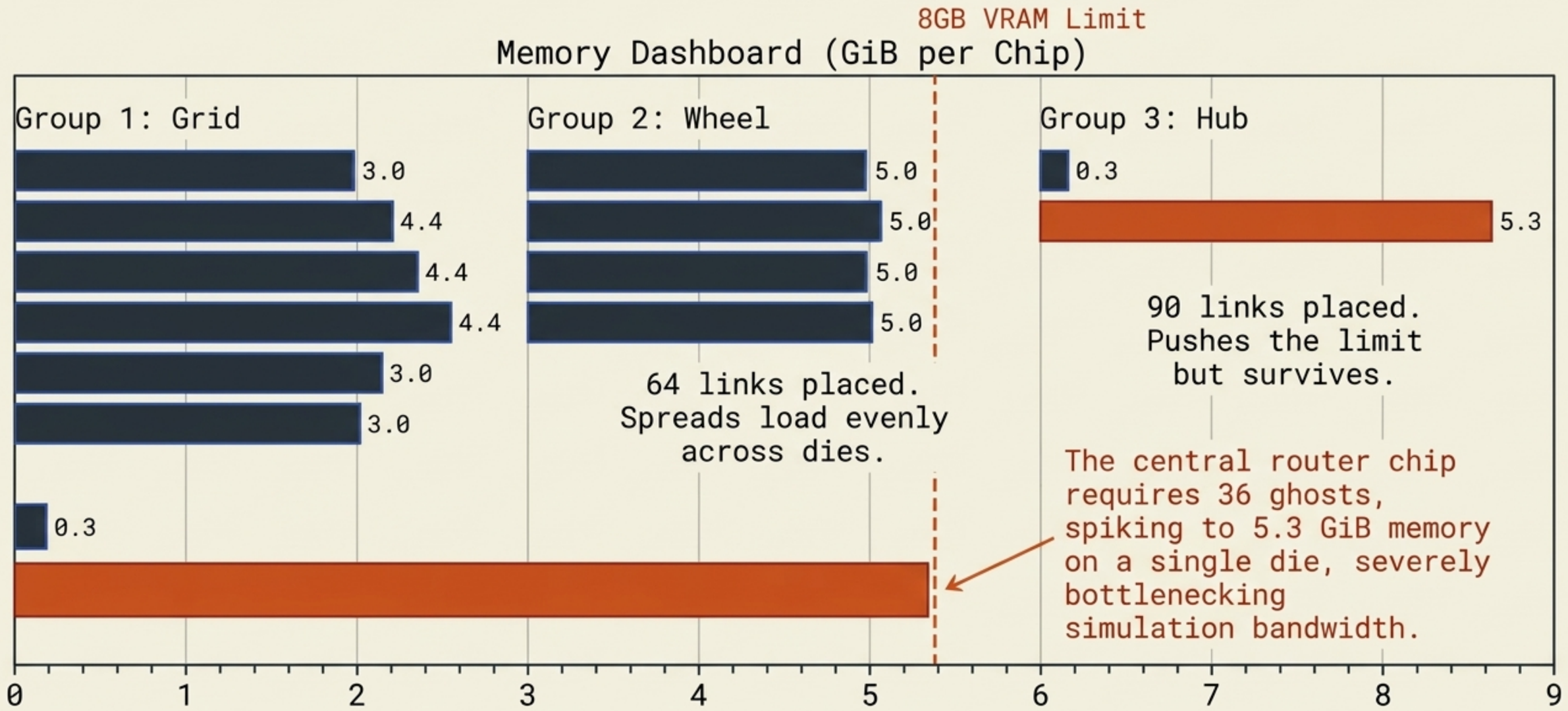


All 15 chip pairs connected. 90 links. 1.00 mean hops (Most connected).

Gemini Notebook

The Topography of Cost: Ghost Loom Profiling

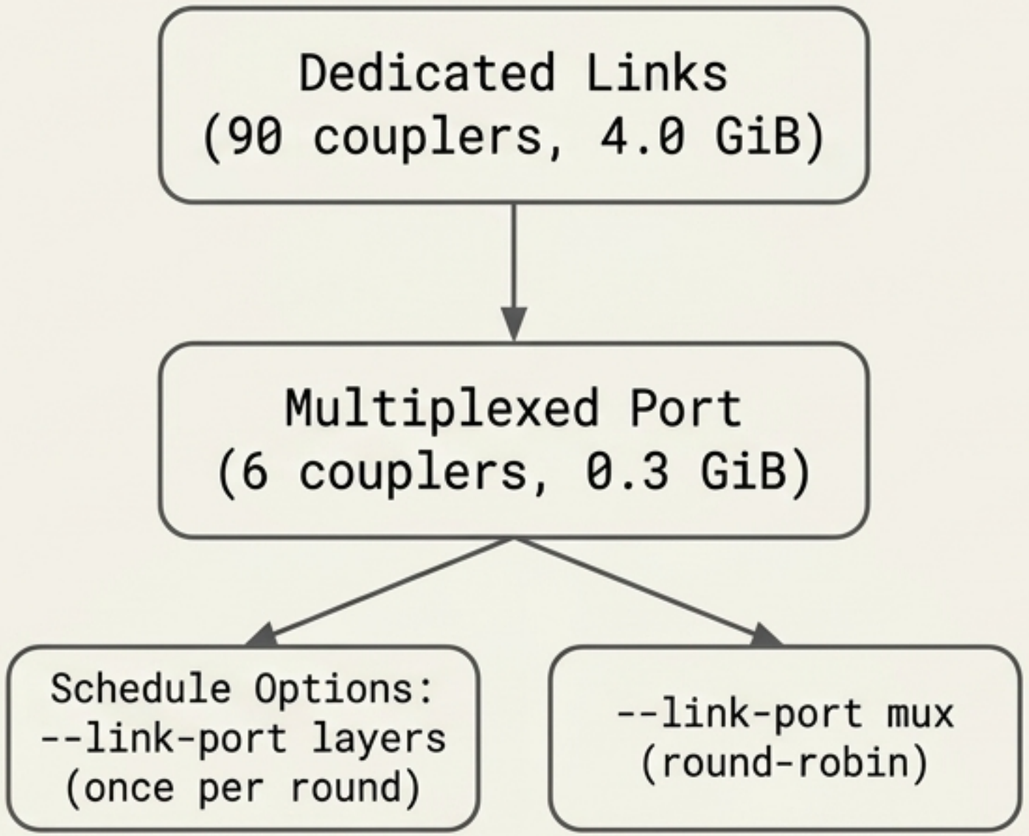
Simulating the hop penalty requires six independent 8 GB dies.

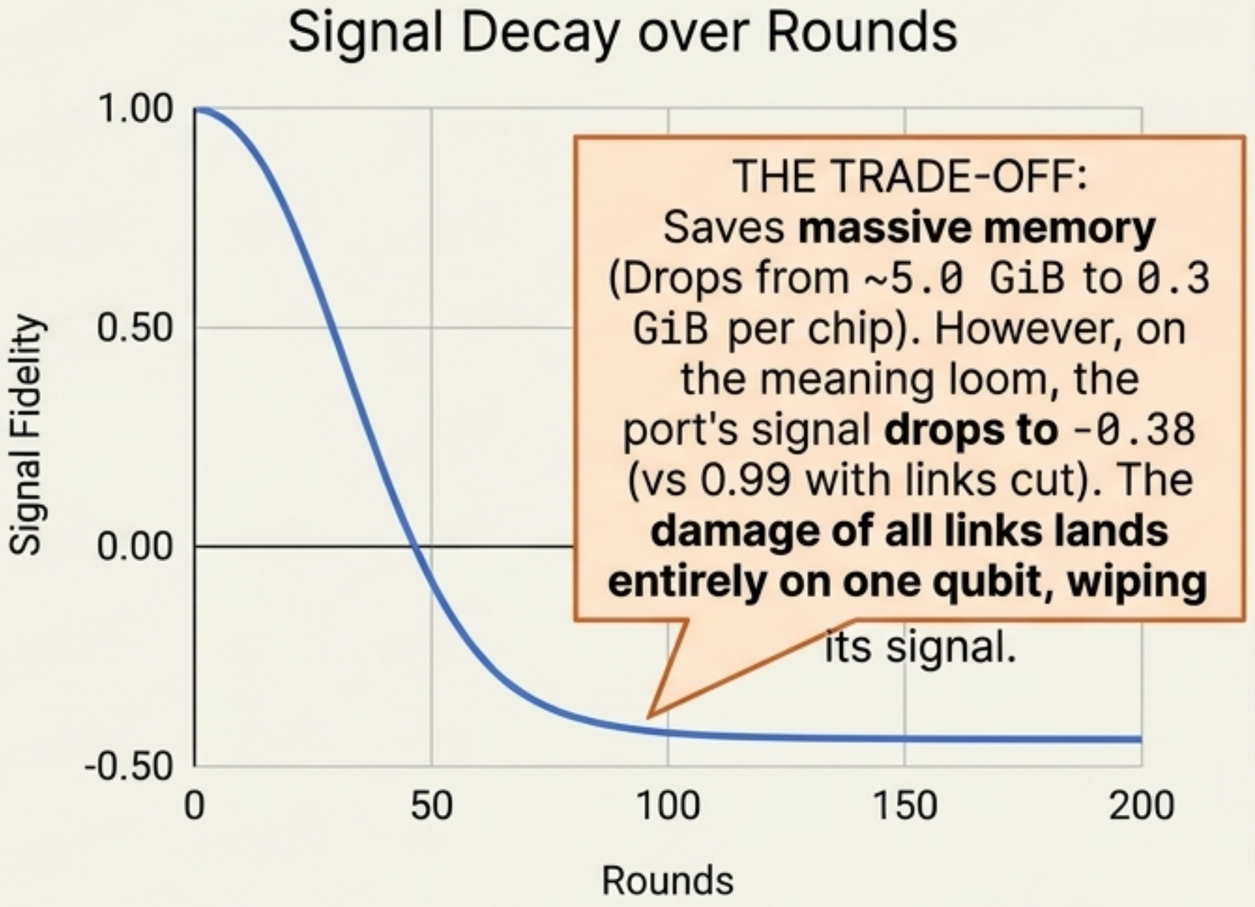


Gemini Notebook

Multiplexing: The Link Port Strategy

Using one router qubit per chip **with a coupler to every partner**, instead of point-to-point l-couplers.





Gemini Notebook

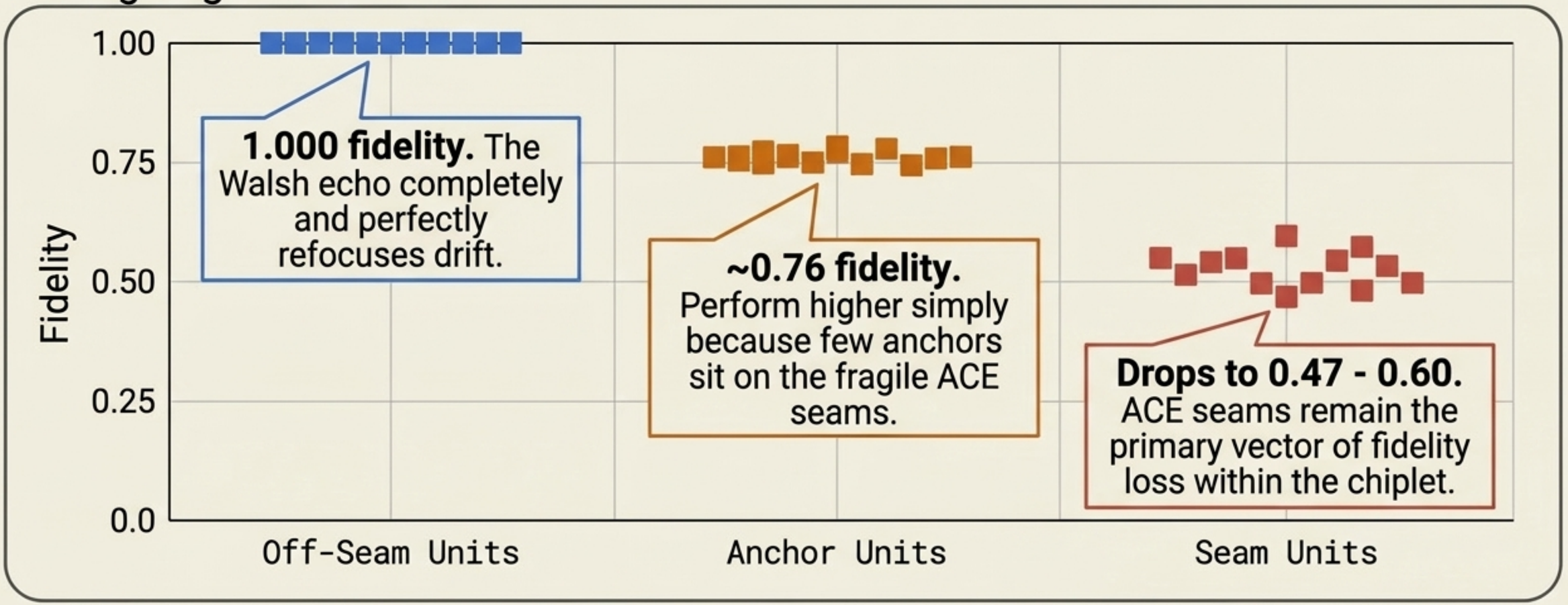
Simulation Engines Head-to-Head

Metric/Feature	Ghost Loom	Mean-Field Loom	Links Cut	QrackBridge
Cross-chip Fidelity Depth 8	0.75 Plateau	0.49	0.49	1.00 (if bonds = couplers)
Entanglement Reach	1 Hop	Correlation only, no entanglement	0	Exact within patches
Memory Cost	Requires 6x 8GB dies	Low	Baseline	Scales aggressively with bonds and depth
Verdict	Leads classical cross-link fidelity.	Virtually identical to cutting links.	Baseline reference.	Perfect for local patches, high memory overhead.

Gemini Notebook

Lab Benchmarks: GHZ Across the Lattice

3-qubit GHZ chains, 1024 shots per setting, simulated across every r3 chiplet using `nighthawk120`.



Gemini Notebook

Error Mitigation: Shor's Code & Syndromes

Cut Between Blocks (1 Link)

Z_L survives any interconnect. Ghost Loom yields 1.000 raw/decoded.

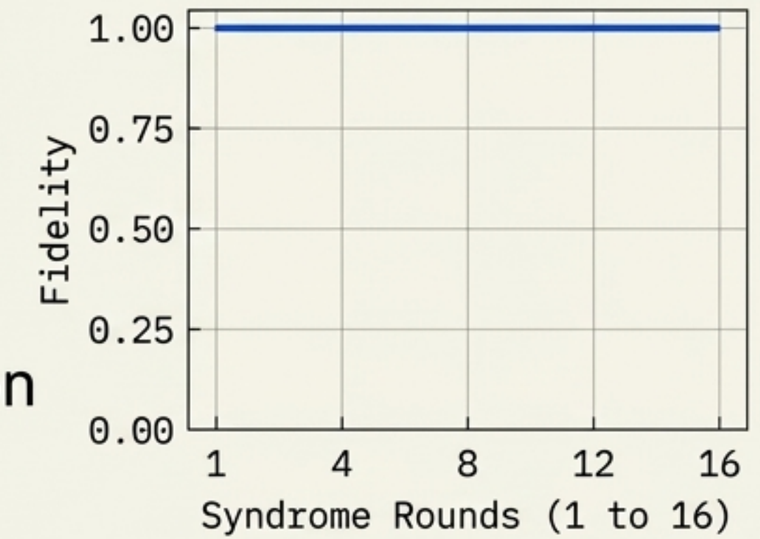
Cut Through Blocks (3 Links)

X_L survives only after majority vote decoding.

QrackBridge Performance

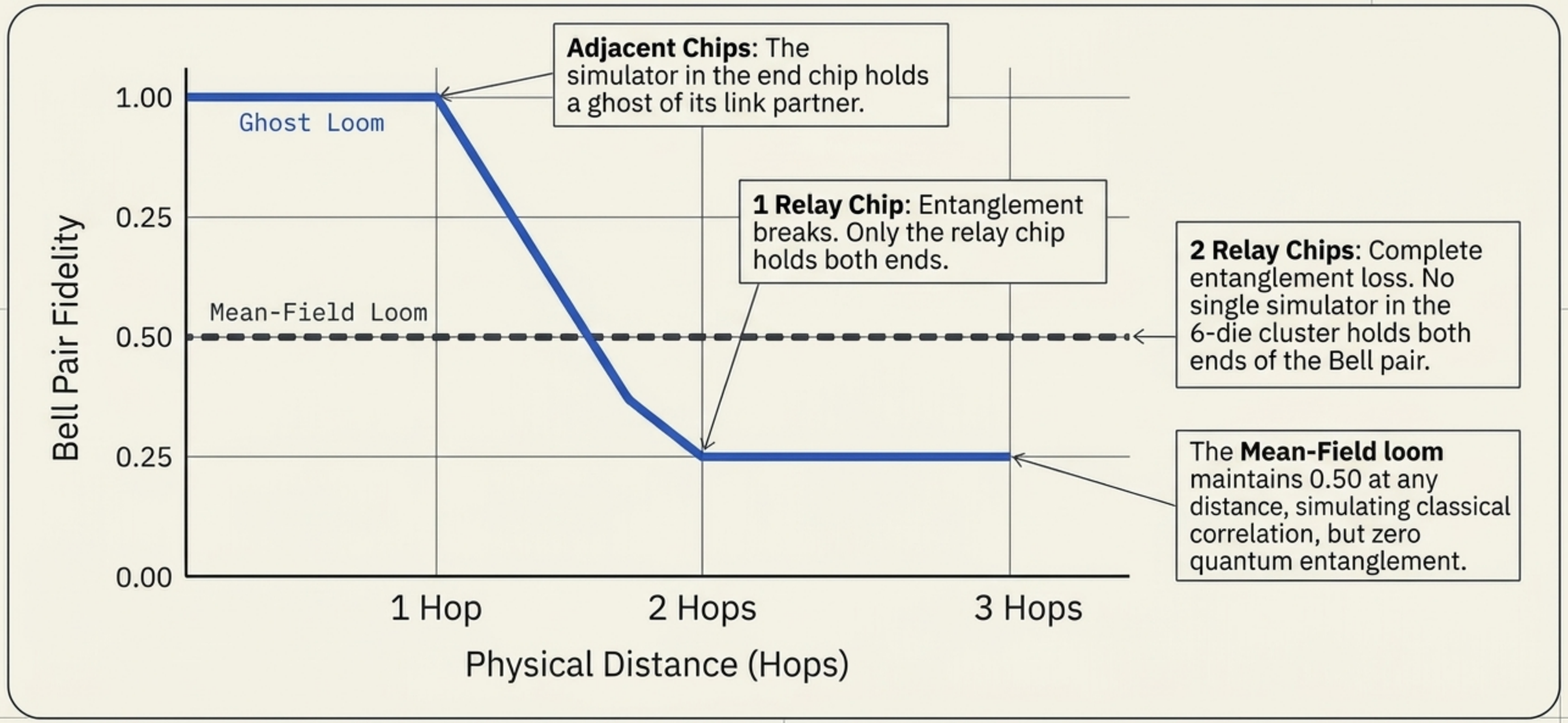
Maintaining logical states across the seam requires 2 to 4 bond qubits.

Achieves perfect 1.00 Z_L / X_L fidelity over 1 to 16 continuous syndrome extraction rounds (measuring known Pauli errors).



Gemini Notebook

Quantum NUMA: The Entanglement Horizon



Gemini Notebook

Quick Reference & Deployment Validation

```
git clone https://github.com/BlueQubitDev/rcs-nighthawk
python3 nighthawk_qrack.py field plan --device r3
python3 nighthawk_qrack.py field selftest
```

✓ Selftests: 14/14 core tests pass (70s CPU).

✓ Dependencies: Requires PyQrack 2.30.6+, Qiskit 2.6.2.

✓ Engine Note: QrackBridge now draws from PyQrack 2.30 C++ native RNG. Seed hashing fixes implemented (bridge_seed()). Shots are strictly distribution-matched, not bit-identical to older Python bridges.

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